



BSE Portal

Internal Operations

USER MANUAL & WORKFLOW GUIDE

How the portal works, screen by screen.

A stock-broking operations portal that stays instant even when the exchange feed is slow or down — with live updates and fully dynamic, per-organisation access control.

 Butter #ffefb3  Forest #013e37

Part A — Mock BSE API · Part B — Internal Portal · v1.0 · 16 Jul 2026

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1 • What BSE Portal is

BSE Portal is an internal web application for a stock-broking firm. It brings together two very different data sources and presents them to staff through role-appropriate screens:

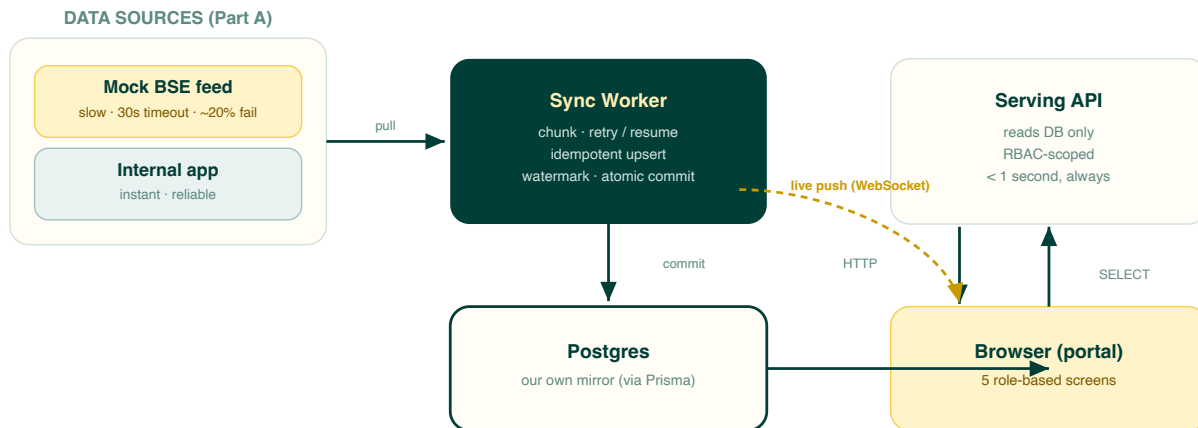
- **The BSE exchange feed** — client master data and trade records. In the real world this feed is *painful*: a full pull takes 5–10 minutes, the network kills any request after 30 seconds, and roughly one pull in five fails partway through. Part A of this project (**Mock BSE API**) faithfully simulates that behaviour.
- **The firm's internal application** — employees and the employee ↔ client mapping (which relationship manager looks after which client). This source is fast and reliable.

From BSE trades combined with the internal mappings, the portal computes **relationship-manager incentives** (a percentage of the brokerage earned on a manager's mapped clients' trades).

The promise. Every screen loads in under a second — even if BSE is completely down — and open screens update on their own when fresh data arrives. No manual refresh, ever.

2 · How it works — the workflow

The trick behind the promise is a clean split: **ingestion** and **serving** never touch each other. A background worker slowly mirrors BSE into our own database; the portal only ever reads that database. So the slow, unreliable feed can never make a screen slow or unavailable.



What happens, step by step

- 1 Ingest continuously.** The sync worker pulls BSE in small pages, each finishing well under the 30-second limit. The instant internal endpoints are pulled every cycle.
- 2 Survive failures.** If a page times out or dies mid-transfer, the worker retries it with backoff and resumes from the last saved position — never from the start, and never committing half a page.
- 3 Never duplicate.** Rows are keyed by their exchange id and upserted, so retries and overlapping refreshes can't create duplicates or contradictions. The rows and the resume position are saved together in one transaction.
- 4 Serve instantly.** Screens read only from Postgres — typically 8–48 ms — with no path to BSE, so they stay fast even if the feed is offline.
- 5 Update live.** The moment new rows are committed, the server pushes a WebSocket event and any open screen refreshes itself.

3 • Getting started

You need **Node.js 20+** and **Docker** (for Postgres). From the project root:

```
npm install          # install all workspaces
npm run db:up        # start Postgres (Docker)
npm run prisma:migrate # create the database schema
npm run seed         # load roles, features & demo users
npm run dev          # start mock API + portal server + web UI
```

Once running, open the portal in your browser:

PORTAL (WEB)

localhost:5173

PORTAL API

localhost:4002

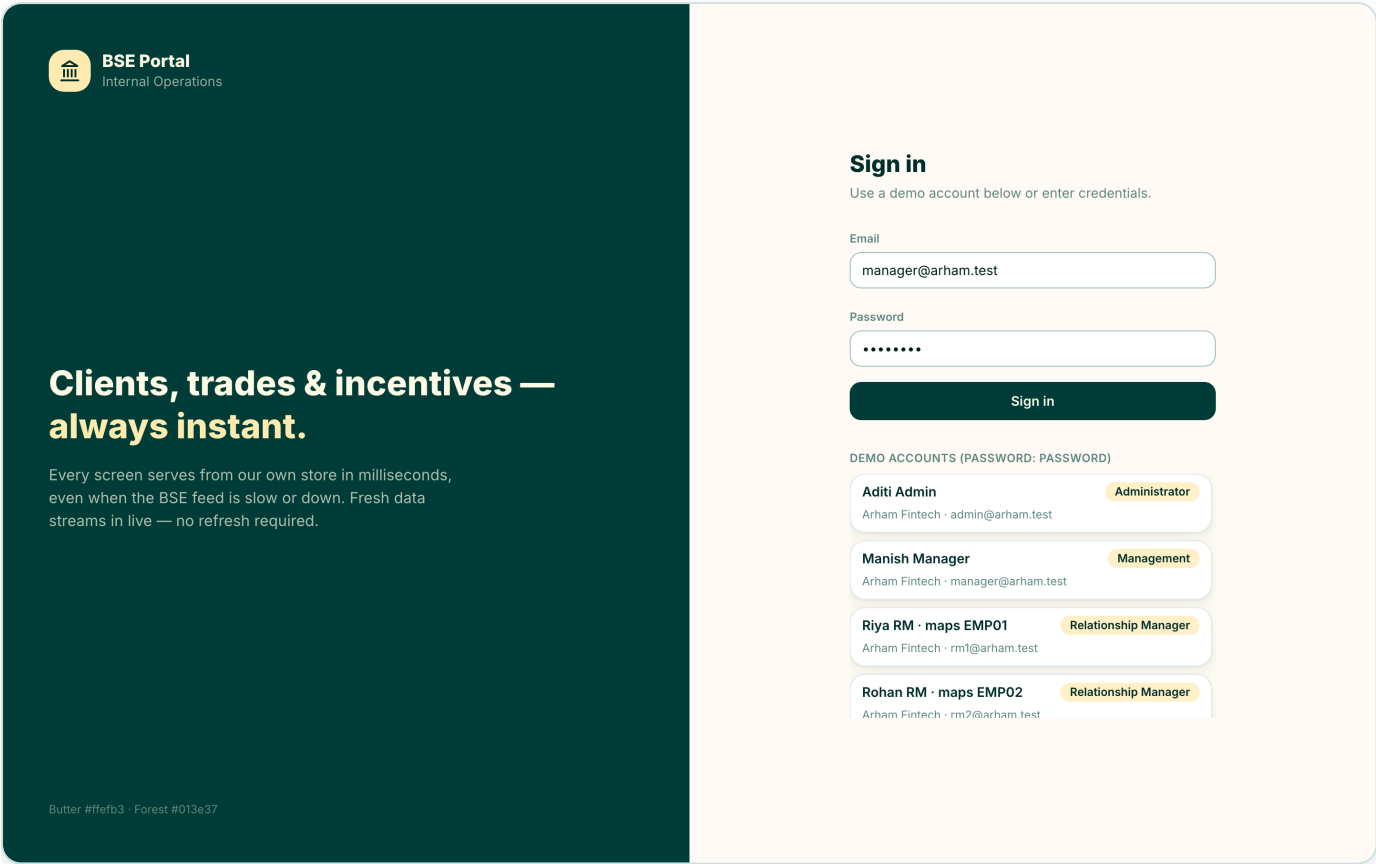
MOCK BSE API

localhost:4001

Tip. On first launch the worker begins mirroring BSE. Watch the clients and trades counts in the top bar climb — that is the background ingestion filling your local store.

4 · Signing in & roles

The sign-in screen lets you type credentials or pick a ready-made demo account. Every demo account uses the password **password**. Your account's **organisation** and **role** decide which screens you see and how much data each screen shows.



The sign-in screen, with one-click demo accounts across two organisations.

How access is decided

Access is a combination of three things — **Organisation × Role × Feature** — and each grant carries a **scope** that narrows the rows you can see:

SCOPE	MEANING	EXAMPLE
ALL	every row	Management sees all clients & all incentives
OWN	only the signed-in user's own records	An RM sees only their own incentive
MAPPED	only rows tied to the user's mapped clients	"My Clients" and that RM's trades

5 · The screens — a guided tour

Clients — feature: clients · scope **ALL**

All clients with key details: PAN, city, trading segment, KYC status, their relationship manager, and when the account was opened. The badge on the right shows the server response time — proof the screen met the sub-second requirement.



BSE Portal

Internal Operations

Clients

ALL

Trades

ALL

Employees

ALL

Incentives

ALL

Manish Manager

Arham Fintech · Management

Sign out

400 clients6,300 trades● Live+ Simulate BSE trades

Clients

• 27 ms

All clients with key details

CLIENT	PAN	CITY	SEGMENT	KYC	RELATIONSHIP MANAGER	OPENED
Reyansh Gupta CL000001	NCBQN5748L	Delhi	EQUITY	VERIFIED	Kabir Joshi	10 Mar 2024
Aditya Patel CL000002	XXES7818X	Chennai	COMMODITY	VERIFIED	Ira Kulkarni	15 Aug 2024
Ira Menon CL000003	MANRH7248E	Kolkata	CURRENCY	VERIFIED	Priya Das	21 May 2024
Kabir Banerjee CL000004	FJGNT3419H	Pune	FNO	VERIFIED	Riya Shah	29 May 2026
Vivaan Pillai CL000005	PIIXC4709E	Ahmedabad	EQUITY	VERIFIED	Arjun Shah	18 Dec 2025
Myra Joshi CL000006	IKKTZ4893F	Delhi	CURRENCY	VERIFIED	Kabir Joshi	05 Nov 2024
Kabir Rao CL000007	HCJXA6463U	Kolkata	FNO	PENDING	Ira Kulkarni	23 Nov 2025
Krishna Banerjee CL000008	SKBRD8017K	Delhi	EQUITY	VERIFIED	Riya Chopra	05 Apr 2026
Krishna Shah CL000009	PZPTF6414V	Ahmedabad	COMMODITY	VERIFIED	Vihaan Das	28 Jun 2024
Saanvi Joshi CL000010	PF0YC6501S	Hyderabad	COMMODITY	VERIFIED	Kiara Joshi	30 May 2026

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Clients — management view. Note the response-time badge and the live data counts in the top bar.

Trades — feature: trades · filterable by client & date

Every trade with symbol, side, quantity, price, value and brokerage. Filter by a specific client id and/or a date range. Management sees all trades; a relationship manager sees only their mapped clients' trades.

BSE Portal

Internal Operations

👤

Clients

ALL

📄

Trades

ALL

👤

Employees

ALL

₹

Incentives

ALL

Manish Manager

Arham Fintech · Management

Sign out

400 clients6,300 tradesLive+ Simulate BSE trades

Trades

Filterable by client and date range

Client ID

From

To

Clear filters

TRADE	CLIENT	SYMBOL	SIDE	QTY	PRICE	VALUE	BROKERAGE	TRADED
TRD00006300	Ira Verma CL000043	ASIANPAINT	BUY	297	₹2,906.15	₹8,63,127	₹341.37	16 Jul, 09:40 pm
TRD00006299	Rohan Iyer CL000072	BHARTIARTL	SELL	211	₹1,437.91	₹3,03,399	₹117.43	16 Jul, 09:40 pm
TRD00006298	Navya Das CL000041	SBIN	SELL	379	₹792.32	₹3,00,289	₹121.24	16 Jul, 09:40 pm
TRD00006297	Arjun Nair CL000262	ITC	SELL	443	₹415.61	₹1,84,115	₹59.83	16 Jul, 09:40 pm
TRD00006296	Navya Chopra CL000270	BAJFINANCE	BUY	369	₹6,588.62	₹24,31,201	₹779.53	16 Jul, 09:40 pm
TRD00006295	Arjun Kulkarni CL000075	TCS	BUY	464	₹4,005.27	₹18,58,445	₹840.37	16 Jul, 09:40 pm
TRD00006294	Ira Banerjee CL000337	KOTAKBANK	SELL	258	₹1,723.37	₹4,44,629	₹178.19	16 Jul, 09:40 pm
TRD00006293	Reyansh Shah CL000020	HDFCBANK	BUY	371	₹1,563.78	₹5,80,162	₹228.52	16 Jul, 09:40 pm
TRD00006292	Aadhya Nair CL000353	WIPRO	BUY	142	₹544.03	₹77,252	₹38.22	16 Jul, 09:40 pm
TRD00006291	Ishaan Mehta	ASIANPAINT	SELL	210	₹1,437.91	₹3,03,399	₹117.43	16 Jul, 09:40 pm

Trades — with client and date-range filters.

Employees — feature: employees · scope ALL

Everyone on the platform, their title and department, and how many clients each relationship manager carries.

BSE Portal

Internal Operations

👤

Clients

ALL

📄

Trades

ALL

👤

Employees

ALL

₹

Incentives

ALL

Manish Manager

Arham Fintech · Management

Sign out

400 clients6,300 tradesLive+ Simulate BSE trades

Employees

All employees on the platform

EMPLOYEE

TITLE

DEPARTMENT

EMAIL

MAPPED CLIENTS

Vihaan Das EMP01	Relationship Manager	Sales	vihaan.das@arhamfintech.ai	26
Myra Shah EMP02	Relationship Manager	Sales	myra.shah@arhamfintech.ai	31
Riya Chopra EMP03	Relationship Manager	Sales	riya.chopra@arhamfintech.ai	35
Vihaan Sharma EMP04	Relationship Manager	Sales	vihaan.sharma@arhamfintech.ai	25
Kiara Joshi EMP05	Relationship Manager	Sales	kiara.joshi@arhamfintech.ai	22
Riya Shah EMP06	Relationship Manager	Sales	riya.shah@arhamfintech.ai	34
Karan Menon EMP07	Relationship Manager	Sales	karan.menon@arhamfintech.ai	24
Priya Das EMP08	Relationship Manager	Sales	priya.das@arhamfintech.ai	32
Kabir Joshi EMP09	Relationship Manager	Sales	kabir.joshi@arhamfintech.ai	28
Ira Kulkarni EMP10	Relationship Manager	Sales	ira.kulkarni@arhamfintech.ai	31
Ira Gupta EMP11	Relationship Manager	Sales	ira.gupta@arhamfintech.ai	24
Ishaan Sharma				

Employees — synced from the internal application.

Incentives — feature: incentives · ALL for management, OWN for an RM

Incentive = 15% of the brokerage on each relationship manager's mapped clients' trades. Management sees every RM with totals up top; an RM sees only their own row.

BSE Portal

Internal Operations

Clients

ALL

Trades

ALL

Employees

ALL

₹ Incentives

ALL

Manish Manager

Arham Fintech · Management

Sign out

400 clients6,300 tradesLive+ Simulate BSE trades

Incentives

All relationship managers · 15% of brokerage on mapped clients' trades

RELATIONSHIP MANAGERS

15

TOTAL BROKERAGE

₹17,23,479

on mapped clients' trades

TOTAL INCENTIVE

₹2,58,522

at 15%

RELATIONSHIP MANAGER	MAPPED CLIENTS	TRADES	BROKERAGE	RATE	INCENTIVE
Priya Das EMP08 · Relationship Manager	32	527	₹1,54,672.24	15%	₹23,200.84
Ira Kulkarni EMP10 · Relationship Manager	31	489	₹1,51,172.18	15%	₹22,675.83
Myra Shah EMP02 · Relationship Manager	31	491	₹1,47,810.43	15%	₹22,171.56
Riya Chopra EMP03 · Relationship Manager	35	591	₹1,47,137.88	15%	₹22,070.68
Riya Shah EMP06 · Relationship Manager	34	527	₹1,38,632.91	15%	₹20,794.94
Kabir Joshi EMP09 · Relationship Manager	28	438	₹1,26,519.40	15%	₹18,977.91
Vihaan Das EMP01 · Relationship Manager	26	440	₹1,17,531.79	15%	₹17,629.77
Priya Nair EMP15 · Relationship Manager	29	429	₹1,12,355.49	15%	₹16,853.32
Ira Gupta EMP11 · Relationship Manager	24	383	₹1,01,248.49	15%	₹15,187.27
Vihaan Sharma					

Incentives — management sees all relationship managers, ranked, with firm-wide totals.

My Clients & the scoped RM view

A relationship manager signs in to a deliberately smaller portal: no firm-wide Clients, Employees, or Access Control in the sidebar. "My Clients" shows only the clients mapped to them.

 **BSE Portal**
Internal Operations

Trades
MAPPED

My Clients
MAPPED

₹ Incentives
OWN

Riya RM - maps EMP01
Arham Fintech - Relationship Mana...

Sign out

400 clients6,300 tradesLive+ Simulate BSE trades

My Clients

Clients mapped to you

CLIENT	PAN	CITY	SEGMENT	KYC	OPENED
Krishna Shah CL000009	PZPTF6414V	Ahmedabad	COMMODITY	VERIFIED	28 Jun 2024
Kabir Kulkarni CL000049	YJZBH6859A	Ahmedabad	CURRENCY	VERIFIED	13 Oct 2025
Aadhya Banerjee CL000050	ITZFC90310	Ahmedabad	EQUITY	VERIFIED	22 Mar 2024
Saanvi Menon CL000062	AGQR88034K	Chennai	EQUITY	VERIFIED	20 Apr 2024
Ananya Pillai CL000073	CBGXA7597S	Kolkata	COMMODITY	VERIFIED	05 Feb 2024
Meera Verma CL000102	WMJVL3979L	Kolkata	COMMODITY	VERIFIED	15 Apr 2026
Aarohi Shah CL000108	MLWFG8310R	Bengaluru	EQUITY	VERIFIED	01 Dec 2025
Aadhya Patel CL000116	DUBTY14460	Chennai	EQUITY	VERIFIED	19 Nov 2025
Diya Pillai CL000121	PWSNW8916F	Ahmedabad	FNO	VERIFIED	21 Sept 2025
Krishna Rao CL000123	QPWRH9055Q	Chennai	EQUITY	VERIFIED	16 Mar 2026

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My Clients — a relationship manager sees only their mapped book (scope **MAPPED**).

 **BSE Portal**
Internal Operations

Trades
MAPPED

My Clients
MAPPED

₹ Incentives
OWN

Riya RM - maps EMP01
Arham Fintech - Relationship Mana...

Sign out

400 clients6,300 tradesLive+ Simulate BSE trades

Incentives

Your Incentive · 15% of brokerage on mapped clients' trades

RELATIONSHIP MANAGERS

1

TOTAL BROKERAGE

₹1,17,532

on mapped clients' trades

TOTAL INCENTIVE

₹17,630

at 15%

RELATIONSHIP MANAGER	MAPPED CLIENTS	TRADES	BROKERAGE	RATE	INCENTIVE
Vihaan Das EMP01 - Relationship Manager	26	440	₹1,17,531.79	15%	₹17,629.77

Incentives for the same RM — scoped to **OWN** : just their single row.

6 · Access Control — dynamic RBAC

Administrators get an **Access Control** screen: a live matrix of Organisation × Role × Feature. Nothing here is hardcoded — every screen a user sees is resolved from this matrix at request time, so changing a cell changes that role's access on their very next action.

 **BSE Portal**
Internal Operations

 Clients
ALL

 Trades
ALL

 My Clients
MAPPED

 Employees
ALL

 Incentives
ALL

 **Access Control**
ALL

Aditi Admin
Arham Fintech · Administrator

Sign out

400 clients6,300 tradesLive+ Simulate BSE trades

Access Control

Dynamic Org × Role × Feature matrix — edits apply immediately

Every screen a user sees is resolved from this matrix at request time. Change a cell and that role's access changes on their next action — nothing is hardcoded.
Scopes: **ALL** every row · **OWN** only their own · **MAPPED** only their mapped clients.

Arham Fintech /arham

ROLE	CLIENTS	TRADES	MY CLIENTS	EMPLOYEES	INCENTIVES	ACCESS CONTROL
Administrator admin	ALL	ALL	MAPPED	ALL	ALL	ALL
Management manager	ALL	ALL	— none —	ALL	ALL	— none —
Relationship Manager rm	— none —	MAPPED	MAPPED	— none —	OWN	— none —

Zenith Securities /zenith

ROLE	CLIENTS	TRADES	MY CLIENTS	EMPLOYEES	INCENTIVES	ACCESS CONTROL
Administrator admin	ALL	ALL	MAPPED	ALL	ALL	ALL
Compliance Officer compliance	ALL	ALL	— none —	ALL	— none —	— none —
Relationship Manager rm	— none —	— none —	MAPPED	— none —	OWN	— none —

Tip: sign in as admin@arham.test in one tab and an RM in another — change the RM's grants here and watch their sidebar change on next navigation.

Access Control — two organisations with intentionally different matrices. The same "Relationship Manager" role grants different features in Arham vs Zenith, proving the model is data-driven.

Each cell is a dropdown; pick a scope (or "none" to revoke). The change is saved immediately.

 **BSE Portal**
Internal Operations

👤 Clients
ALL

📄 Trades
ALL

🎯 My Clients
MAPPED

🏢 Employees
ALL

₹ Incentives
ALL

🔒 Access Control
ALL

Aditi Admin
Arham Fintech · Administrator

Sign out

400 clients6,300 tradesLive+ Simulate BSE trades

Access Control

Dynamic Org x Role x Feature matrix — edits apply immediately

Every screen a user sees is resolved from this matrix at request time. Change a cell and that role's access changes on their next action — nothing is hardcoded.
Scopes: **ALL** every row · **OWN** only their own · **MAPPED** only their mapped clients.

Arham Fintech /arham

ROLE	👤 CLIENTS	📄 TRADES	🎯 MY CLIENTS	🏢 EMPLOYEES	₹ INCENTIVES	🔒 ACCESS CONTROL
Administrator admin	ALL	ALL	MAPPED	ALL	ALL	ALL
Management manager	— none —	ALL	— none —	ALL	ALL	— none —
Relationship Manager rm	OWN	MAPPED	MAPPED	— none —	OWN	— none —

Zenith Securities /zenith

ROLE	👤 CLIENTS	📄 TRADES	🎯 MY CLIENTS	🏢 EMPLOYEES	₹ INCENTIVES	🔒 ACCESS CONTROL
Administrator admin	ALL	ALL	MAPPED	ALL	ALL	ALL
Compliance Officer compliance	ALL	ALL	— none —	ALL	— none —	— none —
Relationship Manager rm	— none —	— none —	MAPPED	— none —	OWN	— none —

Tip: sign in as admin@arham.test in one tab and an RM in another — change the RM's grants here and watch their sidebar change on next navigation.

Setting a grant's scope from the dropdown.

Try it. Open the portal as an admin in one tab and as an RM in another. Change the RM role's *Trades* grant to "none", then navigate in the RM's tab — the Trades screen disappears.

7 · Staying fast & live under a failing feed

Always under a second, even if BSE is down

Because screens read only from our own store, they never wait on the exchange. You can stop the Mock BSE API entirely and the portal keeps serving every screen instantly — it simply stops receiving new data until the feed returns. The response-time badge on each screen makes this visible.

Live updates without refreshing

Use the “+ **Simulate BSE trades**” button in the top bar to inject fresh trades into the mock feed. On the next sync cycle the worker ingests them and pushes a WebSocket event; the Trades and Incentives screens update on their own. The **Live** indicator and the counts in the top bar reflect the flow.

Correct under failure

- **Timeouts** — each pull is abandoned at 30s and retried, never left hanging.
- **Mid-pull failures** — a page that dies partway is retried and resumed from the last saved position.
- **Duplicates** — rows are keyed by exchange id and upserted, so retries and overlapping pulls are harmless.
- **Overlapping refreshes** — cycles never overlap, and incremental pulls use a high-water mark so re-reading the boundary is safe.

Behind the scenes. In the server log you'll see the story play out: `x mid-pull failure` → `⤴ retry` → `✓ synced`.

That is the resilience working as designed, not an error.

8 • Reference

Demo accounts (password: `password`)

EMAIL	ORGANISATION	ROLE	SEES
admin@arham.test	Arham Fintech	Administrator	everything, incl. Access Control
manager@arham.test	Arham Fintech	Management	Clients, Trades, Employees, Incentives (all)
rm1@arham.test	Arham Fintech	Relationship Manager	My Clients, own Trades, own Incentive
rm2@arham.test	Arham Fintech	Relationship Manager	a different mapped book
admin@zenith.test	Zenith Securities	Administrator	everything
compliance@zenith.test	Zenith Securities	Compliance Officer	Clients, Trades, Employees (no incentives)
rm@zenith.test	Zenith Securities	Relationship Manager	My Clients + own Incentive (no Trades screen)

Ports

SERVICE	URL
Internal Dashboard (web)	http://localhost:5173
Portal API (+ WebSocket at /ws)	http://localhost:4002
Mock BSE API	http://localhost:4001
Postgres	localhost:5433

Useful configuration

SETTING	EFFECT
<code>MOCK_BSE_DELAY_MS</code>	per-page delay of the mock feed (set high to simulate the 10-minute pull)
<code>MOCK_BSE_FAILURE_RATE</code>	probability a page fails mid-pull (default 0.2)
<code>SYNC_INTERVAL_MS</code>	how often the worker runs a sync cycle
<code>INCENTIVE_RATE</code>	RM share of brokerage (default 0.15)